

PRACTICAL MST - 1

Spotify 2023 Music Analytics using Power BI

Report



Subject Name: Business Intelligence & Tools
Subject Code: 24CSH-341

Submitted By-

Name: Yashika Dagar

UID: 24BAI70566

Submitted To-

Name: Ashiq Nazir Bhat

1. Aim

To analyse the **Spotify 2023 music dataset using Power BI** and develop meaningful insights into song characteristics, streaming popularity, artist involvement, platform presence, and trends over time.

2. Objectives

The objectives of this experiment are:

1. To explore the audio characteristics of popular songs.
 2. To compare the presence and performance of songs across different music streaming platforms.
 3. To analyse the relationship between artist involvement and song success.
 4. To identify temporal trends in song popularity and musical characteristics.
 5. To investigate the cross-platform presence of songs.
 6. To create interactive dashboards and visualizations using Power BI.
 7. To derive analytical findings that can support decision-making in the music industry.
-

3. Tools Required

- Microsoft Power BI Desktop
 - Microsoft Excel – for the Spotify 2023 dataset
 - Spotify 2023.xlsx dataset
 - Power Query – for data preparation and transformation
 - DAX (Data Analysis Expressions) – for calculated measures and columns
 - Computer/Laptop with Windows operating system
-

4. Theory

Power BI

Power BI is a business intelligence and data visualization tool developed by Microsoft. It allows users to import, clean, transform, analyse and visualize data through interactive reports and dashboards.

Power Query

Power Query is used for **data cleaning and transformation** before analysis. It can be used to change data types, remove errors, handle missing values, rename columns and prepare datasets for visualization.

DAX

DAX is a formula language used in Power BI to create **measures, calculated columns and calculated tables**. In this experiment, DAX was used to calculate metrics such as total songs, total streams, average streams, average audio features and platform presence.

Data Visualization

Data visualization represents numerical information through charts and graphs. In this experiment, bar charts, column charts, line charts, scatter plots, cards and slicers were used to identify patterns in the Spotify dataset.

Audio Features

The dataset contains several characteristics describing songs, including:

- **Danceability** – indicates how suitable a song is for dancing.
- **Energy** – represents the intensity and activity level of a song.
- **Valence** – indicates the musical positivity or mood of a song.
- **Acousticness** – represents the extent to which a song has acoustic characteristics.
- **Instrumentalness** – indicates the likelihood of a track being instrumental.
- **Liveness** – represents the presence of live-performance characteristics.
- **Speechiness** – indicates the presence of spoken-word elements.

Cross-Platform Analysis

Cross-platform analysis examines how songs are represented across different services such as **Spotify, Apple Music, Deezer and Shazam**. It helps identify the extent to which popular songs have a presence across multiple platforms.

5. Procedure

Step 1: Import the Dataset

Step 2: Prepare the Data

Step 3: Create DAX Measures

Step 4: Create the Overview Dashboard

Step 5: Music Analysis

Graph 1: Top 10 Most Streamed Songs

Graph 2: Average Audio Characteristics of Popular Songs

Graph 3: Danceability vs Spotify Streams

Graph 4: Energy vs Spotify Streams

Step 6: Platform Comparison

Step 7: Artist Impact Analysis

Step 8: Temporal Trend Analysis

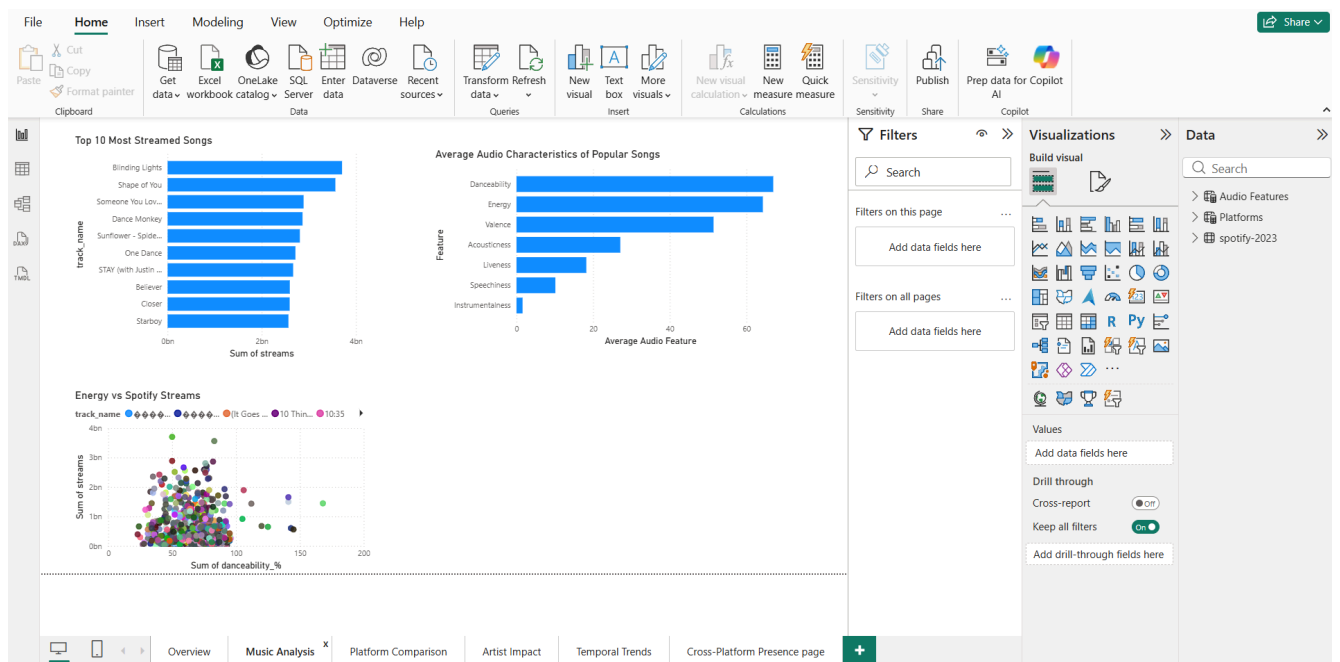
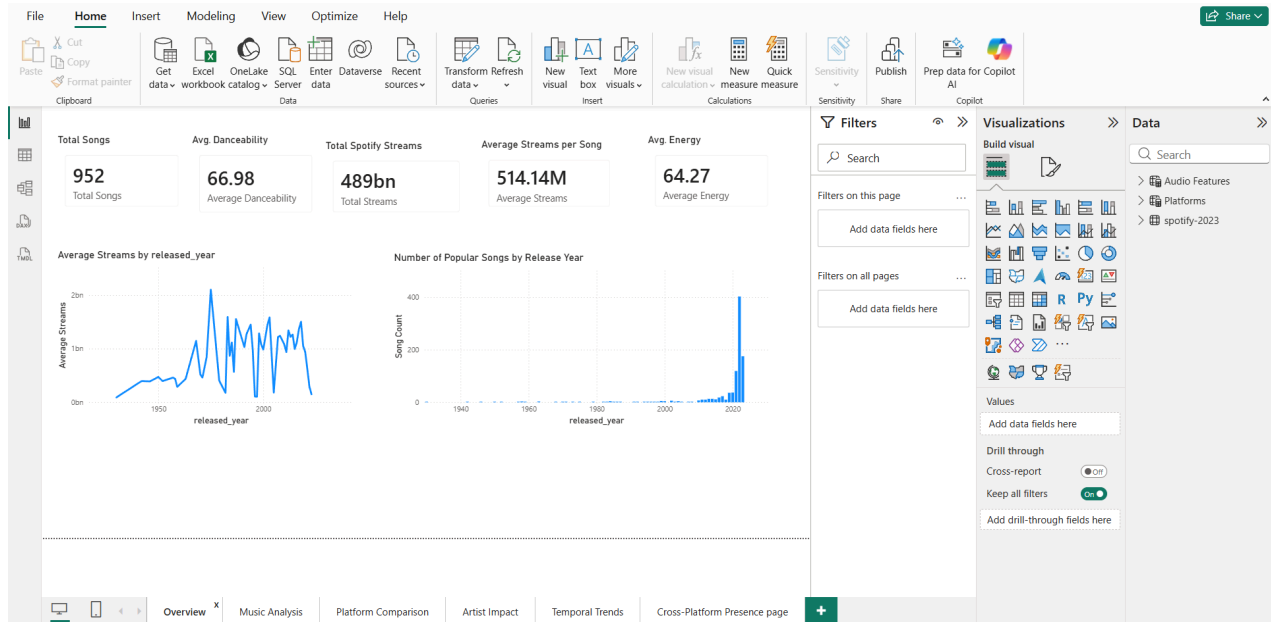
Step 9: Cross-Platform Presence Analysis

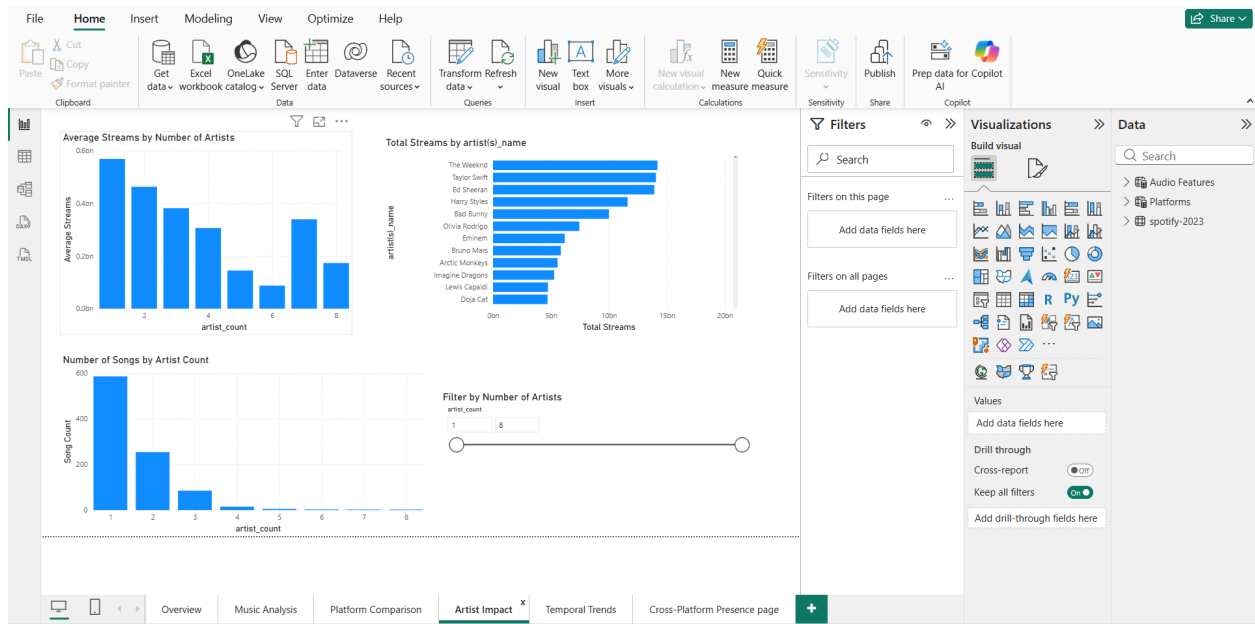
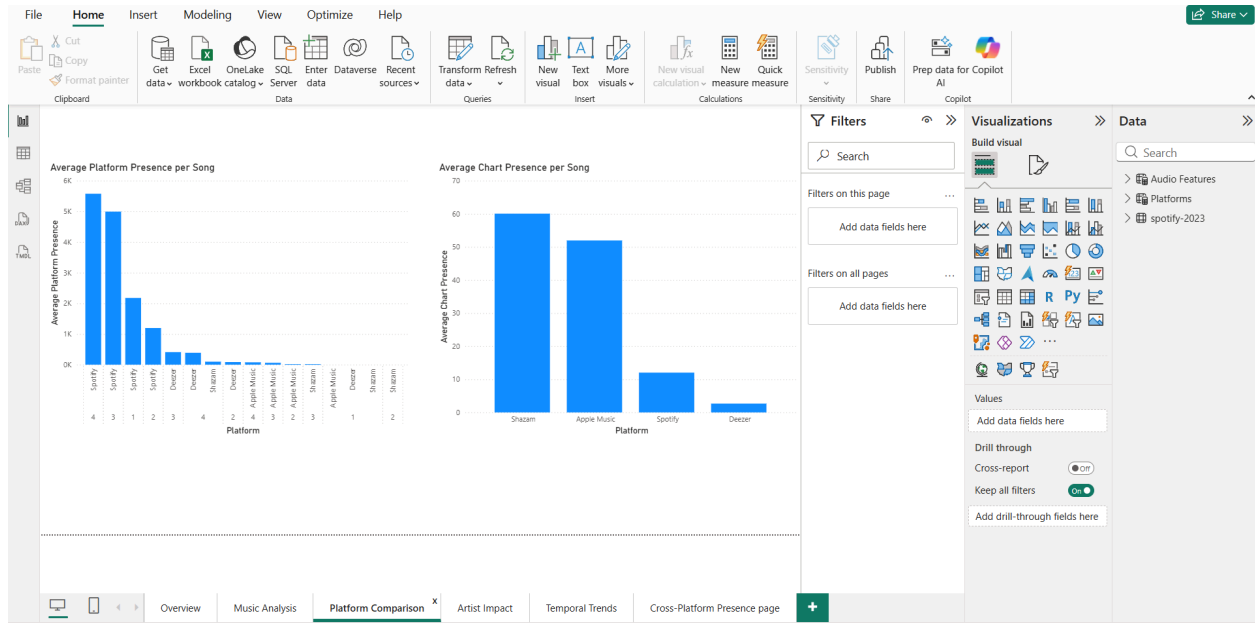
6. Result

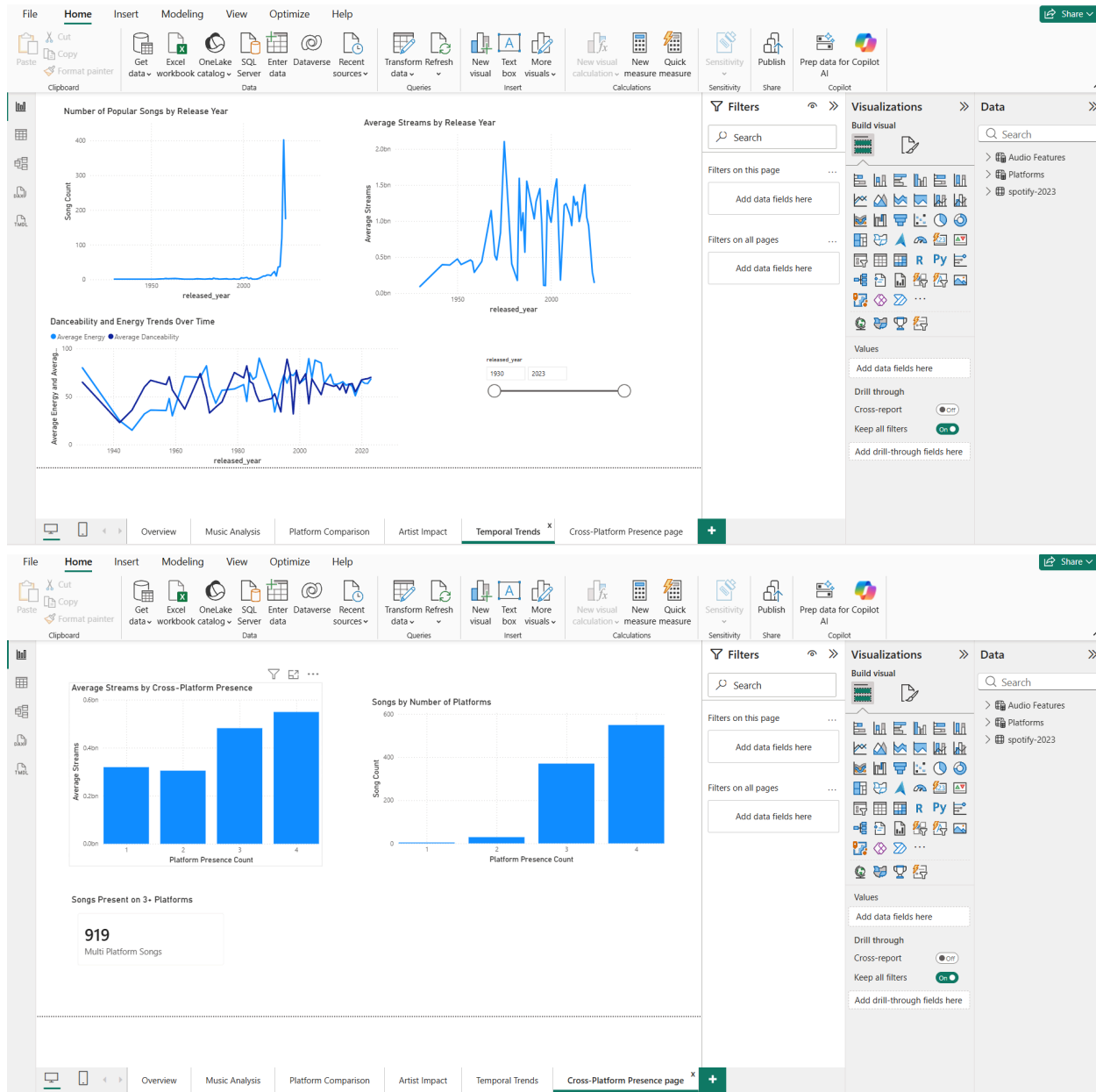
The Spotify 2023 dataset was successfully analysed using **Microsoft Power BI**. Interactive dashboards were developed to examine **music characteristics, streaming popularity, artist involvement, platform presence and temporal trends**.

The analysis provides a multidimensional view of popular songs by combining audio features with streaming and platform-related information. The visualizations allow users to compare highly streamed songs, examine audio characteristics, investigate artist collaborations, observe

changes across release years, and evaluate the cross-platform presence of songs. The resulting Power BI report provides an interactive analytical framework that can be used to understand patterns in popular music and support data-driven analysis in the music industry.







7. Learning Outcomes

After completing this experiment, I was able to:

1. Understand the structure and characteristics of a real-world music dataset.
2. Import and prepare Excel data in Microsoft Power BI.
3. Create DAX measures and calculated columns for analytical calculations.
4. Analyse audio features such as danceability, energy, valence and acousticness.



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

**NAAC
GRADE A+**

Accredited University

5. Compare streaming and platform-related metrics across different music services.
6. Analyse the relationship between artist involvement and song popularity.
7. Identify temporal patterns using release-year-based visualizations.
8. Perform cross-platform analysis using calculated platform presence.
9. Create interactive dashboards using charts, KPI cards and slicers.
10. Interpret data visualizations and derive meaningful insights from a real-world dataset.
11. Present analytical findings in a structured Power BI report for decision-making in